

AI-Powered Precision Oncology: Cancer Drug Response Prediction and Recommendation System

[Report]

Introduction :

Cancer treatment effectiveness varies significantly across different cancer types and drugs. Predicting drug sensitivity can help researchers identify effective treatments and support precision oncology. This project aims to predict cancer drug response using machine learning and deep learning techniques on the Genomics of Drug Sensitivity in Cancer (GDSC) dataset and develop a drug recommendation system.

Objective :

- Predict whether a cancer cell line is sensitive or resistant to a drug.
- Compare the performance of multiple machine learning and deep learning models.
- Identify the most influential factors affecting drug response.
- Develop a drug recommendation engine based on predicted sensitivity.

Dataset Description :

The project uses the GDSC dataset containing 242,035 observations and 19 variables. Each observation represents the response of a cancer cell line to a specific anti-cancer drug.

Column	Description
COSMIC_ID	Unique identifier of the cancer cell line
CELL_LINE_NAME	Name of the cancer cell line
TCGA_DESC	Cancer type classification
DRUG_ID	Unique identifier of the drug
DRUG_NAME	Name of drug tested
LN_IC50	Log-transformed IC50 value measuring drug sensitivity
AUC	Area Under Dose-Response Curve
Z_SCORE	Standardized drug response
GDSC Tissue	Broad tissue category
Cancer Type	Specific cancer
Microsatellite	Genetic instability status
Growth Properties	Adherent/Suspension growth
CNA	Copy Number Alteration data available
Gene Expr	Gene expression data available
Methylation	DNA methylation data available
TARGET	Drug target protein
TARGET_PATHWAY	Biological pathway targeted by drug

Target Variable :

The original dataset provides LN_IC50, which measures the concentration of a drug required to inhibit 50% of cancer cell growth.

- Lower LN_IC50 values indicate higher drug effectiveness.
- Higher LN_IC50 values indicate lower drug effectiveness.

To formulate a classification problem, a new variable called Response was created:

- Sensitive (1): LN_IC50 below the median value.
- Resistant (0): LN_IC50 above the median value.

This transformation created a balanced classification dataset suitable for machine learning and deep learning models.

Data Preprocessing :

- Missing values in categorical variables were replaced with "Unknown".
- Duplicate records were checked and removed where necessary.
- Data leakage was identified and prevented by removing LN_IC50, AUC, and Z_SCORE from the predictor variables.
- Categorical features were encoded using one-hot encoding for traditional machine learning models.
- Native categorical handling was used for CatBoost.

Exploratory Data Analysis :

Several visualizations were performed to understand the dataset:

- Distribution of LN_IC50
- Drug response distribution
- Top 10 cancer types
- Top 10 Most tested drugs
- Drug effectiveness across cancer types
- Average IC50 by Drug
- Proportion of Drug Response by Top Cancer Types

The analysis showed substantial variation in drug response across cancer types and drug categories.

Model Development :

The following models were trained and evaluated:

1. Logistic Regression
2. Random Forest
3. XGBoost

4. CatBoost
5. Artificial Neural Network

Performance was evaluated using Accuracy and ROC-AUC scores.

Results :

Model	Accuracy	ROC-AUC
Logistic Regression	82.29%	0.9105
Random Forest	81.50%	0.8983
XGBoost	81.13%	0.9036
CatBoost	83.45%	0.9216
Neural Network	83.24%	0.9183

CatBoost achieved the best overall performance with an accuracy of 83.45% and ROC-AUC of 0.9216.

Feature Importance Analysis :

Feature importance analysis revealed:

- Drug Name was the most influential predictor (51.5% importance).
- Tissue descriptors significantly influenced treatment response.
- Growth properties and drug targets contributed meaningfully to predictions.
- Target pathways provided additional biological insight.

These findings suggest that drug characteristics and cancer tissue origin play a major role in determining treatment effectiveness.

Explainable AI :

SHAP analysis was used to interpret model predictions and understand the contribution of individual features. The analysis confirmed that drug identity, tissue descriptors, growth properties, and cancer type were the primary drivers of drug sensitivity predictions.

Drug recommendation system:

A drug recommendation engine was developed using the final CatBoost model. For a given cancer profile, the model evaluates all candidate drugs and estimates the probability of achieving a sensitive response.

The system ranks drugs according to their predicted sensitivity probabilities and recommends the highest-ranked candidates. For the selected cancer profile, Dactinomycin, Docetaxel, Sepantronium bromide, Paclitaxel, and Rapamycin emerged as the top recommended drugs, each with predicted sensitivity probabilities above 94%.

This demonstrates how machine learning can be used to prioritize potentially effective therapeutic options and support precision oncology research. The recommendation engine extends the project beyond simple classification by providing actionable drug rankings based on model predictions.

Conclusion :

This project developed an AI-powered Cancer Drug Response Prediction and Recommendation System using the GDSC dataset. Multiple machine learning and deep learning models were evaluated, including Logistic Regression, Random Forest, XGBoost, CatBoost, and Neural Networks. After removing data leakage and performing comprehensive model evaluation, CatBoost achieved the best performance with an accuracy of 83.45% and a ROC-AUC score of 0.9216. Feature importance and SHAP analysis identified drug identity, tissue descriptors, and cancer characteristics as key factors influencing drug response. A drug recommendation engine was also developed to rank drugs based on predicted sensitivity probabilities, demonstrating the potential of machine learning in precision oncology and cancer treatment research.

Future work :

Future work may include incorporating detailed genomic data, integrating additional cancer datasets, exploring advanced deep learning models such as Graph Neural Networks, and developing a Streamlit-based web application for real-time drug recommendations. Further validation on independent datasets and patient-level clinical data could improve the robustness and practical applicability of the system.